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REVIEW ARTICLE

Digital twin for neurological conditions: a systematic scoping review

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Abstract
Digital twin technology has emerged as a promising approach for advancing precision medicine in complex diseases, particularly in neurology. This scoping review aimed to systematically map the current literature on digital twin applications in neurological conditions, focusing on study characteristics, modeling approaches, data sources, and clinical applications. A comprehensive search across six databases from inception to February 2026 identified 36 studies that met the inclusion criteria. The findings demonstrated a rapid increase in publications since 2023, with Alzheimer's disease ($n=14$) and stroke ($n=12$) representing the most frequently studied conditions. Data-driven and machine learning–based models were the most common approaches, followed by mechanistic and hybrid models. Major application domains included clinical trial optimization, disease progression modeling, risk prediction, and personalized treatment. However, most studies were proof-of-concept or retrospective in design, with limited external validation and no randomized controlled trials identified. Additionally, many models lacked continuous real-time data integration, indicating that few met the criteria of fully dynamic digital twins. These findings suggest that while digital twins hold substantial potential to transform precision neurology, significant challenges remain in terms of methodological rigor, standardization, and clinical translation. Future research should prioritize prospective validation, integration of multimodal real-world data, and the development of standardized frameworks to support safe and effective implementation.

Keywords Neurological conditions · Digital twin · In silico · Brain health · Digital health

1 Introduction

Neurological disorders represent one of the leading causes of disability and mortality worldwide, accounting for an increasing proportion of the global disease burden due to population aging and improved survival from other chronic conditions [1, 2]. Major neurological conditions, including Alzheimer's disease, Parkinson's disease, multiple sclerosis, epilepsy, and stroke, are characterized by complex pathophysiological mechanisms involving multiscale interactions between genetic, molecular, structural, and functional factors [3, 4]. These disorders often exhibit substantial inter-individual heterogeneity in disease progression, treatment response, and clinical outcomes, making precision medicine approaches particularly challenging [5]. Traditional neurological research has largely relied on population-level statistical models, which may fail to adequately capture individual variability and dynamic disease trajectories [6].

Recent advances in artificial intelligence (AI), computational neuroscience, and biomedical data science have enabled new paradigms for modeling complex diseases through personalized computational representations; among these, the concept of the digital twin has emerged as a promising approach for integrating multimodal patient data into dynamic computational models that simulate disease progression and predict treatment responses [7, 8]. Originally developed in engineering and manufacturing, digital twin

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technology refers to a virtual representation of a physical entity that is continuously updated with real-world data to enable simulation, monitoring, and optimization [9]. In healthcare, this concept has evolved into the notion of human digital twins—computational models that integrate clinical, biological, behavioral, and environmental data to support personalized diagnosis and treatment strategies [10].

In neurology, digital twins are particularly relevant given the complexity of the human brain and the growing availability of multimodal datasets, including neuroimaging, electrophysiological recordings, genomics, and digital biomarkers [11]. The concept of a "virtual brain twin" has been proposed as a personalized computational model that integrates structural and functional brain data to simulate neural dynamics and disease processes [12]. These models aim to provide mechanistic insights into neurological diseases while enabling individualized predictions regarding disease progression and therapeutic outcomes [13]. For example, digital twin approaches have been explored to model network dysfunction in epilepsy, simulate neurodegenerative processes in Alzheimer's disease, and predict disability progression in multiple sclerosis [12, 14].

One of the major potential advantages of digital twins in neurological conditions lies in the ability to move beyond descriptive models toward predictive and actionable frameworks. By integrating longitudinal patient data, digital twins may allow clinicians to perform in silico testing of interventions, optimize treatment strategies, and identify early disease markers before clinical deterioration becomes apparent [15]. Furthermore, digital twin models may facilitate virtual clinical trials by enabling simulations of treatment effects in patient-specific contexts, potentially reducing the time and cost of drug development [16].

Digital twin approaches also contribute to early detection and risk stratification in neurodegenerative diseases. For instance, recent studies have demonstrated the feasibility of deriving digital biomarkers from computational brain models using electroencephalographic data to predict Alzheimer's disease progression and cognitive decline [17]. Similarly, personalized computational models integrating neuroimaging data have shown promise in predicting disease trajectories and supporting treatment decision-making in chronic neurological disorders [18]. These developments suggest that digital twins could become an important component of future precision neurology frameworks.

Despite these promising developments, the implementation of digital twin technology in neurological care remains in its early stages. Several challenges continue to limit widespread adoption, including data integration, model validation, interpretability, ethical considerations, and regulatory oversight [10, 19]. Neurological diseases present additional complexities due to the brain's multiscale organization, nonlinear dynamics, and limited accessibility of direct biological measurements compared to other organ systems [12]. Furthermore, the development of reliable digital twins requires large-scale longitudinal datasets, standardized data pipelines, and robust validation frameworks, which are still evolving in neurological research [20].

Another important consideration is the distinction between digital twins and related computational concepts such as digital models, simulations, and virtual cohorts. While many studies use these terms interchangeably, true digital twins require dynamic bidirectional data integration between physical and virtual systems [9]. Recent scoping reviews suggest that only a minority of studies fully meet the strict definition of a digital twin, highlighting the need for clearer conceptual frameworks and standardized terminology in the field [21]. Establishing consistent definitions is particularly important for neurological applications, where computational modeling approaches vary widely in scope, complexity, and clinical readiness.

Although several narrative reviews and technical papers have examined digital twin technologies in healthcare, the literature specifically focusing on neurological applications remains fragmented. Existing reviews often focus on specific diseases, technological components, or theoretical frameworks rather than providing a comprehensive overview of how digital twin approaches are being applied across neurological conditions [12, 22]. Furthermore, rapid advances in AI, computational modeling, and digital health technologies have led to a growing body of literature that has not yet been systematically mapped.

Scoping reviews are particularly well-suited to emerging and interdisciplinary research areas such as digital twins in neurology, as this approach allows for the identification of key concepts, research gaps, and future directions without restricting inclusion to narrowly defined study designs [23]. By systematically mapping the current evidence base, scoping reviews can help clarify how digital twin technologies are conceptualized, developed, and applied across neurological conditions. This method may also help identify methodological trends, technological enablers, and translational barriers that may influence future research and clinical implementation.

Therefore, the objective of this scoping review is to systematically map the current literature on digital twin applications in neurological conditions. Specifically, this review aims to (1) identify the neurological disorders in which digital twin approaches have been applied, (2) characterize the types of data and modeling approaches used, (3) summarize reported clinical and research applications, and (4) identify current limitations and research gaps. By providing a comprehensive overview of the existing literature, this review

seeks to inform future research directions and support the development of clinically meaningful digital twin frameworks for neurological diseases.

2 Methods

2.1 Study design

This review was conducted in accordance with the Arksey and O'Malley framework and further refined by subsequent methodological advancements [24]. A scoping review approach was deemed appropriate given the emerging and heterogeneous nature of digital twin applications in neurological conditions, where evidence spans diverse study designs, computational methodologies, and clinical contexts [25]. The reporting of this review adhered to the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews) guidelines to ensure methodological transparency and reproducibility [26]. The protocol was registered with the Open Science Framework (https://osf.io/jqsxu/overview).

2.2 Search criteria

A comprehensive and systematic literature search was performed across multiple electronic databases, including PubMed, Embase, Cochrane Library, CINAHL, ProQuest, and Google Scholar. The search strategy was developed using a combination of controlled vocabulary terms, including Medical Subject Headings (MeSH) and Emtree terms, as well as free-text keywords related to digital twin technologies and neurological diseases. The search strategy was structured around two primary concepts: digital twin technologies and neurological conditions, encompassing disorders such as stroke, Alzheimer's disease, Parkinson's disease, epilepsy, multiple sclerosis, traumatic brain injury, and mental disorders. Boolean operators (AND, OR) were applied to combine search terms, and database-specific adaptations were implemented to optimize sensitivity and specificity. The PubMed search strategy included terms such as "Digital Twins" and "digital twin*" combined with neurological disease-related terms including "neurolog*," "brain," "stroke," "Alzheimer*," "Parkinson*," "epilep*," and "mental disorder*". The search was limited to studies published in English. The search was further carried out from the inception to February 15, 2026, reflecting the rapid advancement of digital twin technologies in healthcare [27]. In addition, the reference lists of included studies and relevant review articles were manually screened to identify any additional eligible publications. Because digital twin research in neurology is an emerging field, full-text preprints and peer-reviewed conference proceedings were also considered when sufficient methodological detail was available and when an original empirical, computational, or methodological contribution was reported.

2.3 Inclusion and exclusion criteria

Eligibility criteria were defined using the Population–Concept–Context (PCC) framework recommended for scoping reviews (Table 1) [26]. The population of interest included individuals with neurological conditions, such as stroke, Alzheimer's disease, Parkinson's disease, epilepsy, multiple sclerosis, traumatic brain injury, and mental disorders. The concept focused on digital twin technologies. Because terminology remains heterogeneous in healthcare and neurological research, digital twin eligibility was operationalized through a tiered definition prior to full-text assessment. A study was considered eligible when it reported an explicit or functionally equivalent digital twin, virtual twin, human digital twin, digital twin trial, virtual brain twin, or closely related patient-, disease-, or system-specific virtual representation, and when the model was linked to neurological, clinical, biological, neurophysiological, imaging, wearable, sensor-derived, trial, or experimentally derived data. Eligible studies also needed to describe a clinical or translational purpose, including individualized simulation, monitoring, disease trajectory prediction, treatment-response estimation, rehabilitation adaptation, clinical trial optimization, risk prediction, or decision support.

Studies were included in the investigation of digital twin or digital twin–adjacent systems applied to neurological conditions that reported original empirical, computational, methodological, framework development, or proof-of-concept findings. Eligible publication types

Table 1 Population, concept, context framework for defining study eligibility criteria

Component	Description
Population	Individuals with neurological conditions (e.g., stroke, Alzheimer's disease, Parkinson's disease, epilepsy, multiple sclerosis, traumatic brain injury, mental disorders)
Concept	Patient-, disease-, or neurological system-specific virtual representations linked to empirical or clinically relevant data for individualized simulation, monitoring, prediction, treatment optimization, rehabilitation adaptation, clinical trial optimization, or decision support; digital twin–adjacent classification for general AI models, static classifiers, and generic simulations
Context	Clinical, research, or healthcare settings involving diagnosis, prediction, monitoring, or treatment

included peer-reviewed journal articles, full-text preprints with sufficient methodological detail, peer-reviewed conference proceedings, methodological framework papers, and conceptual or position papers when an original digital twin architecture, implementation agenda, or methodological contribution was presented. Studies were required to be written in English and to provide sufficient methodological detail for evaluation. Studies were excluded if not involving neurological conditions, lacking any digital twin or equivalent virtual modeling framework, being editorials, commentaries, conference abstracts without full-text methodological detail, general review articles without original methodological or framework contribution, or focusing solely on engineering simulations without clinical or translational relevance [28]. Conventional artificial intelligence classifiers, static prediction models, or generic computational simulations were not interpreted as fully functional digital twins unless a patient-, disease-, or neurological system-specific virtual representation and clinical or translational use were explicitly described. Because digital twin research in neurology remains in its early stages, eligible studies were retained for mapping but subsequently classified by digital twin maturity.

2.4 Data extraction

Data extraction, also referred to as data charting, was conducted using a standardized form developed a priori based on established scoping review methodology. Extracted data included study characteristics such as author, publication year, and country, as well as methodological details including study design, type of neurological condition, and characteristics of the digital twin model. Additional variables included data sources used for model development, such as imaging data, clinical records, or wearable sensor data, as well as reported outcomes, including predictive performance, clinical utility, and implementation feasibility.

Two independent reviewers performed the screening and data extraction processes. Titles and abstracts were initially screened for relevance, followed by full-text reviews of potentially eligible studies. Any discrepancies between reviewers were resolved through discussion, and when necessary, a third reviewer was consulted to reach consensus. Extracted data were synthesized descriptively, focusing on categorizing digital twin applications, identifying methodological trends, and summarizing reported clinical implications. To distinguish digital twin systems from broader computational modeling approaches, additional data were extracted on digital twin-defining features, including patient or disease specificity, empirical data linkage, model instantiation or calibration, evidence of model updating, temporal or longitudinal data integration, bidirectional data flow, simulation or intervention-testing functionality, and intended clinical or translational use. These variables were used to support the operational classification of the included studies by digital twin maturity. Additional synthesis focused on validation strategy, interpretability, scalability, clinical readiness, data requirements, and translational limitations across different digital twin architectures.

2.5 Operational classification of digital twin maturity

Each included study was categorized according to digital twin maturity. Dynamic prototype digital twins were defined as systems with patient- or system-specific virtual representations, longitudinal or sensor-derived data integration, iterative model updating, and feedback or adaptation functions intended to support individualized decision-making. Prototype or partial digital twins were defined as systems that included patient-, disease-, or system-specific virtual representations and clinically oriented simulation, prediction, or trial-optimization functions but lacked continuous real-time updating, bidirectional data flow, or prospective clinical integration. Conceptual or framework-level digital twins were defined as studies that proposed explicit digital twin architectures, consensus rules, or implementation frameworks without full empirical deployment. Digital twin–adjacent enabling models were defined as computational, sensing, or data-infrastructure models that supported digital twin development but did not satisfy the criteria for fully dynamic digital twins. This classification was used for descriptive synthesis and did not alter the final number of included studies.

2.6 Qualitative methodological appraisal

Although formal quality assessment is not a mandatory component of scoping reviews, a qualitative methodological appraisal was conducted to contextualize the maturity of the evidence rather than to determine eligibility [26]. The appraisal was based on five predefined domains relevant to computational and digital twin studies: (1) study design and empirical basis, (2) clarity of model architecture and input–output specification, (3) validation strategy, (4) clinical or translational readiness, and (5) reporting transparency and reproducibility. These domains were informed by general scoping-review guidance and by digital-twin verification, validation, and uncertainty quantification considerations [26, 29].

A low rating was assigned when a study provided a clearly specified digital twin or virtual modeling architecture, explicit data sources and input variables, quantitative validation or structured clinical/implementation evaluation,

and sufficient methodological detail for interpretation. A medium rating was assigned when a study described a relevant digital twin or digital twin–adjacent model but relied primarily on retrospective, simulation-based, proof-of-concept, conceptual, or internally validated evidence without external validation, prospective validation, or clinical implementation. Studies with insufficient methodological detail to judge model structure, data source, or digital twin relevance were excluded during full-text review. The appraisal was performed independently by two reviewers, and disagreements were resolved through discussion. Ratings were used only for descriptive synthesis and did not influence study inclusion or exclusion.

3 Results

3.1 Study selection results

A total of 332 records were identified through systematic searching across all six electronic databases (PubMed: $n=98$; Embase: $n=72$; Cochrane Library: $n=14$; CINAHL: $n=31$; ProQuest: $n=55$; Google Scholar: $n=62$). After removing 128 duplicate records, 204 unique records underwent title and abstract screening. Of these, 81 records were excluded at the screening stage due to not meeting the minimum eligibility criteria for neurological condition relevance or digital twin modeling components. The remaining 123 records were retrieved for full-text assessment.

After title and abstract screening, 123 database-derived records were retrieved for full-text assessment. In addition, 3 records were identified through reference-list screening, resulting in 126 full-text reports considered for eligibility. During full-text review, 90 reports were excluded for the following reasons: not related to digital twin or an equivalent virtual modeling framework ($n=33$), not involving neurological conditions ($n=38$), non-peer-reviewed publication, editorial, conference abstract, or general review article without original methodological or framework contribution ($n=11$), and non-English language ($n=8$). Following full-text review, 36 studies met all eligibility criteria and were included in the final synthesis (Fig. 1). During full-text assessment, studies were evaluated using the operational digital twin criteria described above. Because the purpose of this scoping review was to map an emerging and heterogeneous field, studies were not excluded solely for lacking continuous real-time updating or bidirectional data flow. Instead, the absence of these features was coded as part of the digital twin maturity classification. This approach allowed the review to distinguish fully dynamic or prototype digital twins from partial, conceptual, and digital twin–adjacent computational models.

Inter-rater agreement for title/abstract screening was $\kappa=0.82$ (95% CI: 0.76–0.88), and for full-text review was $\kappa=0.79$ (95% CI: 0.71–0.87), both indicating substantial agreement. All discrepancies were resolved through discussion without recourse to a third reviewer.

The 36 included studies were published between 2021 and 2026, with a marked increase in publication volume from 2023 onward: 2021 ($n=1$), 2022 ($n=3$), 2023 ($n=5$), 2024 ($n=11$), 2025 ($n=9$), and 2026 ($n=7$). This temporal pattern reflects the rapid expansion of digital twin research in neurological conditions. The studies originated from 12 countries, with the United States contributing the largest number of publications ($n=18$, 50.0%), followed by Canada ($n=3$, 8.3%), Italy ($n=3$, 8.3%), the United Kingdom ($n=3$, 8.3%), Australia ($n=2$, 5.6%), and the Netherlands ($n=2$, 5.6%). The remaining studies originated from Austria, China, Denmark, Pakistan, Sweden, and Switzerland ($n=1$ each). One study was conducted within a multinational consortium involving 12 countries and 19 organizations [30], and was counted across multiple countries; therefore, country-level counts are not mutually exclusive.

3.2 Characteristics of the included studies

The studies encompassed a wide range of neurological conditions, study designs, digital twin approaches, data sources, and application domains, reflecting the methodological heterogeneity characteristic of this emerging field. The 36 included studies are summarized in Table 2. To complement the tabular summaries, Fig. 2 presents the overall distributional patterns of the included studies across neurological conditions, modeling paradigms, data sources, and clinical or research application domains. Bar charts were used because several categories, including neurological condition, data source, and application domain, were not mutually exclusive.

To address heterogeneity in the use of digital twin terminology across literature, Table 2 also reports an operational classification of digital twins for each included study. This classification distinguishes dynamic prototype digital twins, prototype or partial digital twins, conceptual or framework-level digital twins, and digital twin–adjacent enabling models.

Alzheimer's disease was the most frequently studied condition, addressed in 14 studies (38.9%) [17–35, 41, 45, 54–58, 62]. Stroke represented the second most common focus, featuring in 12 studies (33.3%) [30, 33, 35, 37–40, 44, 47, 48, 59, 61]. Amyotrophic lateral sclerosis (ALS) was the subject of 2 studies (5.6%) [52, 53], and multiple sclerosis was investigated in two studies (5.6%) [36, 50]. Individual studies addressed Parkinson's disease [49],

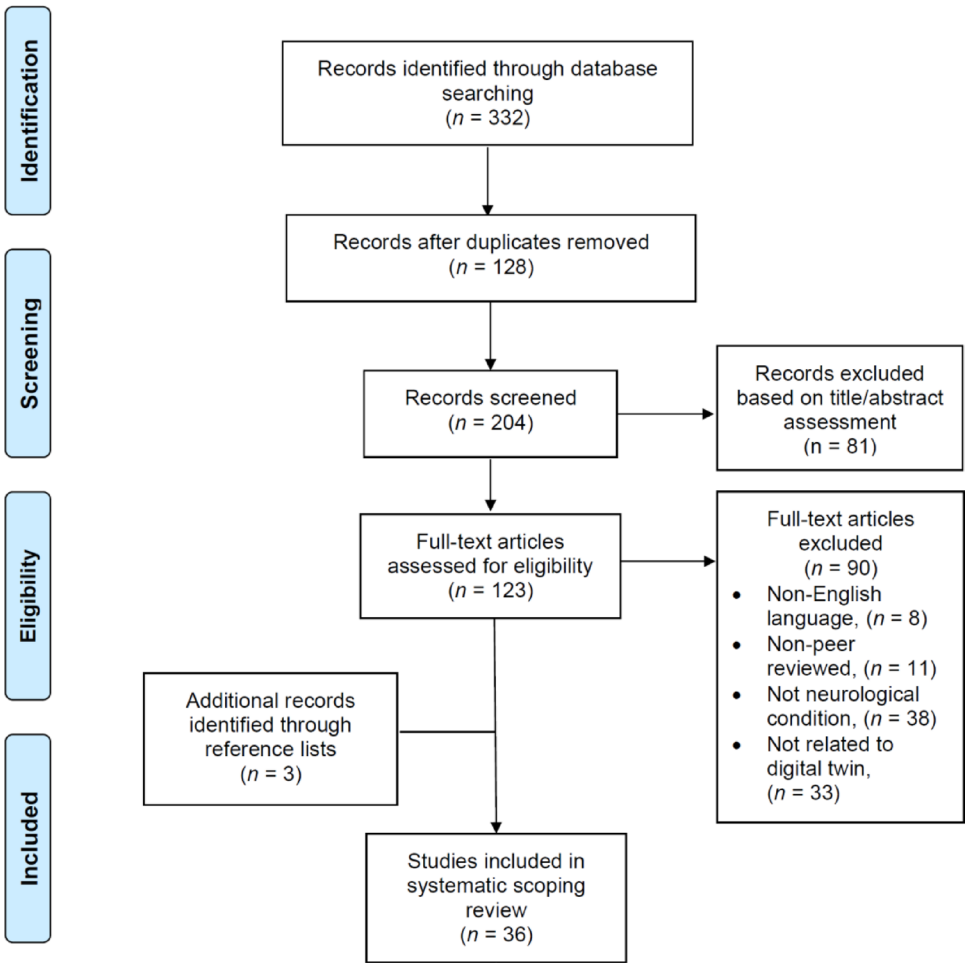


Fig. 1 PRISMA flow chart for the systematic scoping review on digital twins in neurological conditions

epilepsy [39], glioblastoma [43], disorders of consciousness [64], Pompe disease (neuromuscular) [46], schizophrenia [51], dementia [63], and general neurological rehabilitation [42], each contributing one study. Some studies addressed multiple neurological conditions; therefore, counts are not mutually exclusive.

The included studies employed diverse methodological designs and publication formats. Computational modeling and proof-of-concept studies constituted the largest group ($n = 16$, 44.4%), followed by retrospective clinical or simulation studies ($n = 10$, 27.8%), methodological or framework-development studies ($n = 5$, 13.9%), prospective validation studies ($n = 2$, 5.6%), expert consensus studies ($n = 1$, 2.8%), and conceptual framework or position studies proposing digital twin architectures or implementation agendas ($n = 2$, 5.6%). Full-text preprints were retained when an original empirical, computational, or methodological contribution was present, and sufficient methodological detail was available. Peer-reviewed conference proceedings were also retained when full-text methods and results were reported. Conference abstracts without full methodological detail, editorials, commentaries, and general narrative reviews without original methodological or framework contribution were excluded. No randomized controlled trials employing digital twins as primary intervention tools were identified, consistent with the field's early translational stage.

The data sources used for digital twin development varied substantially across studies. Electronic health record (EHR) data and clinical trial datasets were most frequently employed ($n = 14$, 38.9%), followed by neuroimaging data including brain MRI and CT ($n = 9$, 25.0%), electrophysiological recordings including EEG and MEG ($n = 5$, 13.9%), multiomics data ($n = 4$, 11.1%), wearable and sensor data ($n = 3$, 8.3%), patient-derived experimental data ($n = 2$, 5.6%), and expert consensus data ($n = 1$, 2.8%). Several studies integrated multiple data modalities; for example, Funk et al. [41] combined APOE4 genotyping, amyloid and tau biomarkers, neuropsychological scores, hippocampal atrophy measures, and lipidomic profiles within a single computational framework.

Clinical trial optimization was the most frequently reported application domain, identified in 9 studies (25.0%) [30–33, 43, 46, 52, 53, 62]. Disease progression modeling and risk prediction were each represented in 8 studies

Table 2 Summary of methodological characteristics and clinical applications of digital twin research in neurology

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[17]	Italy	To enable early diagnosis of Alzheimer's disease using digital twins combined with non-invasive neurophysiological recordings	Proof-of-concept/validation study	Alzheimer's Disease (preclinical/SCD stage)	Computational brain model driven by neurophysiological data	Prototype digital twin	Non-invasive neurophysiological recordings (EEG/MEG)	Early diagnosis	Digital twins combined with non-invasive recordings enabled detection of AD in the subjective cognitive decline (SCD) stage. The approach overcomes limitations of invasive diagnostic methods.	Low
[31]	Canada	To develop a digital twin technique using external observational data to reduce sample sizes in Alzheimer's disease clinical trials	Methodological/simulation study	Alzheimer's Disease	Statistical/machine learning model	Partial digital twin for clinical trial optimization	External observational study data, clinical trial data	Clinical trial optimization	External observational data-based digital twins significantly reduced required RCT sample sizes. The approach aligns with regulatory frameworks for trial augmentation.	Medium
[32]	Canada	To develop a Digital Twin Trial (DTT) methodology combining cognitive decline prediction with prediction-powered inference for AD RCT sample size reduction	Methodological/simulation study	Alzheimer's Disease	Interpretable machine learning model (DTT methodology)	Partial digital twin trial methodology	Retrospective patient database, 18-month RCT simulations	Clinical trial optimization	DTT methodology reduced variance within treatment groups by adjusting observed scores with digital twin predictions. The approach demonstrated consistent sample size reduction across simulated trials.	Medium
[33]	USA	To apply a variational autoencoder-based digital twin model for forecasting disease progression trajectories in stroke patients	Retrospective computational study	Stroke (Ischemic)	Machine learning model (variational autoencoder)	Partial digital twin trajectory simulation	EHR data (ICD codes, lab values) from pre-stroke patients	Disease progression modeling/ Clinical decision support	Simulated patient trajectories were virtually indistinguishable from real data (AUC adversary = 0.51). The model can support clinical decision-making and virtual control arms.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[34]	USA	To develop a digital twin clinical decision support tool for individualized Alzheimer's prediction and prevention	Tool development/proof-of-concept study	Alzheimer's Disease	Bayesian computational model (Fulcrum BHN model)	Partial digital twin decision-support model	Literature-curated data (1,500 + papers), clinical and observational study datasets	Clinical decision support/Personalized prevention	Digital twins predicted individual MCI and dementia risk and ranked personalized intervention combinations. Clinical prototype was refined through interviews with neurologists and primary care physicians.	Medium
[35]	USA	To develop and validate a proof-of-concept digital twin model for predicting individual treatment outcomes in post-stroke aphasia	Preprint/proof-of-concept retrospective study	Stroke (Post-stroke Aphasia)	Patient-specific continually updating predictive model	Dynamic prototype digital twin	Lesion characteristics, brain integrity, and systemic health data from aphasia patients	Treatment outcome prediction/Personalized therapy	The digital twin predicted individual naming outcomes during language treatment and quantified modifiable risk factor impacts. The approach captures heterogeneous recovery patterns beyond group-level models.	Medium
[36]	Canada	To demonstrate a digital twin approach for modeling onset of disease-specific brain atrophy in multiple sclerosis using MRI	Retrospective computational study	Multiple Sclerosis	Statistical/spline-based modeling	Digital twin-adjacent individualized trajectory model	Brain MRI data (longitudinal and cross-sectional normal aging data)	Disease progression modeling/Precision medicine	The digital twin approach estimated the age of onset of MS-specific brain atrophy from MRI. Mixed spline models identified from 52 covariates accurately characterized individual disease trajectories.	Medium
[37]	USA	To develop DELPHI expert consensus rules for a digital twin model of acute stroke care in the neurocritical care unit	Expert consensus/Delphi study	Stroke (Acute)	AI-based digital twin (rule-based/expert consensus model)	Conceptual/framework-level digital twin	Expert consensus (clinical rules), existing sepsis digital twin model	Critical care management/Clinical decision support	Expert consensus rules for a neurocritical care digital twin were established via Delphi methodology. The framework expands an existing sepsis digital twin model to acute stroke management.	Low

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[38]	UK (international)	To propose a scientific agenda for patient-specific digital twins that dynamically model thrombosis and stroke risk	Conceptual framework/position statement	Stroke (Thrombosis-related)	Hybrid model (physics-based + data-driven)	Conceptual/framework-level digital twin	Multimodal longitudinal data (imaging, lab tests, wearables, EHR) proposed framework	Risk prediction/Personalized anticoagulation therapy	Current clinical scores inadequately capture dynamic thrombosis risk. A unified digital twin framework integrating mechanistic and ML models is proposed for personalized stroke and bleeding risk prediction.	Low
[39]	Netherlands	To develop a digital twin of the epileptic brain network to predict optimal surgical strategies in medically refractory focal epilepsy	Proof-of-concept/pilot study	Epilepsy (focal, medical refractory)	Computational brain network model	Prototype digital twin	Intracranial EEG data (single-pulse electrical stimulation, Engel 1 A outcomes)	Surgical planning/Treatment optimization	Re-analysis of delayed responses to SPES enabled novel computational modelling of epileptic networks. The approach aims to predict surgical outcomes and optimize resection strategy.	Medium
[40]	Italy	To develop a digital twin model of the left atrium for personalized stroke risk assessment in atrial fibrillation patients	Proof-of-concept/computational study	Stroke (AF-related)	Computational fluid dynamics (CFD) model	Prototype digital twin	Patient-specific cardiac imaging data (dynamic LA models)	Risk prediction/Personalized stroke prevention	CFD-based digital twin of the left atrium improved personalized stroke risk assessment beyond CHA2DS2-VASc score. The model captures patient-specific thromboembolic risk in AF patients.	Medium
[41]	USA	To create a brain homeostatic digital twin model for functional assessment of lipidomics in Alzheimer's disease	Computational modeling study	Alzheimer's Disease	Systems biology/mechanistic model (232 functional processes)	Partial mechanistic digital twin	Multiomics data (ADNI: APOE4, Abeta, Tau, MMSE, hippocampal atrophy, lipidomics; $N = 20$)	Biomarker discovery/Disease mechanism understanding	11 lipidomic modules correlated with 11 functional processes in domains of microglia metabolism and neuron energy balance. Digital twins bridged omics data to functional AD processes.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[42]	Austria	To develop an integrated digital twin model for monitoring and predicting therapy progress in neurological patients using ICF-based documentation	Tool development/feasibility study	Neurological disorders (general rehabilitation)	ICF-based data integration model	Partial rehabilitation digital twin	Therapy device data (exoskeletons, physical therapy robotics), standardized outcome measures (Berg Balance Scale, SARA, MAS)	Rehabilitation monitoring/ Therapy progress prediction	The Abilitate app integrated ICF-standards with device data for continuous neurological rehabilitation monitoring. The system enables personalized therapy progress documentation and prediction.	Medium
[43]	UK	To develop a mathematical digital twin model of glioblastoma to design virtual clinical trials for BMP4 differentiation therapy	Preprint/virtual clinical trial simulation study	Glioblastoma (Brain tumor)	Mathematical/ mechanistic model	Prototype digital twin	Experimental data from 12 patient-derived glioma stem cell (GSC) lines	Clinical trial design/Treatment optimization	Continuous BMP4 delivery during radiotherapy was more effective than single dose. Virtual trials showed patient stratification by tumor proliferation rate improved trial success probability.	Low
[44]	Sweden	To develop and validate a hybrid digital twin model combining mechanistic and machine learning approaches for stroke risk simulation	Preprint/ hybrid modeling study	Stroke	Hybrid model (mechanistic + machine learning)	Partial hybrid digital twin	Physiological and biochemical variables from clinical datasets	Risk prediction/ Disease progression modeling	The hybrid digital twin combined mechanistic disease trajectory simulation with ML-based risk estimation. This complementary approach enhanced personalized stroke risk assessment.	Medium
[45]	USA	To provide an identifiability-guided framework for rigorous assessment of digital twins in Alzheimer's disease clinical research and care	Methodological/framework study	Alzheimer's Disease	Data-driven computational model (assessment framework)	Methodological/framework-level digital twin	Clinical AD data (sparse, longitudinal)	Model validation/ Clinical research methodology	Identifiability analysis is critical for reliable digital twin deployment in AD. The framework addresses challenges of sparse and noisy medical data in individual-level predictions.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[46]	USA	To develop a QSP-based digital twin approach to supplement clinical trial data for enzyme replacement therapies in Pompe disease	Computational modeling/in silico clinical trial	Pompe Disease (neuromuscular)	Quantitative Systems Pharmacology (QSP) model	Prototype QSP digital twin	Clinical trial data for enzyme replacement therapy (ERT), mechanistic model parameters	Clinical trial optimization/Drug efficacy comparison	QSP digital twins performed in silico comparison of avalglucosidase alfa vs. standard of care in virtual IOPD patients. The approach addressed limitations of small patient numbers in rare disease trials.	Medium
[47]	Pakistan	To develop an RF sensing and AI-based digital twin model for stroke detection and patient monitoring	Proof-of-concept/technical study	Stroke	Machine learning/deep learning model (Stacked AutoEncoders, KNN)	Digital twin-adjacent sensing and prediction model	Ultra Wide Band (UWB) RF backscattering signals	Stroke detection/Patient monitoring	Fine KNN algorithm achieved 93.4% binary and 92.3% multi-class stroke classification accuracy. The SAE-based feature extraction with Fine KNN yielded best autonomous classification performance.	Medium
[48]	Denmark	To design and implement a human digital twin system for upper-limb stroke rehabilitation using exoskeletons and serious games	Case study	Stroke (Upper-limb rehabilitation)	Human digital twin (HDT) model (ETHICA methodology)	Dynamic prototype human digital twin	Exoskeleton sensor data, patient performance data from serious games	Rehabilitation monitoring/Adaptive therapy	The HDT system enabled real-time therapy difficulty adjustment based on patient condition. The ETHICA methodology provided a structured approach for HDT implementation in stroke rehabilitation.	Medium
[49]	USA	To couple brain network simulation with pharmacokinetics for Parkinson's disease toward patient-usable digital twins	Computational modeling/simulation study	Parkinson's Disease	Brain network simulation + pharmacokinetic (PK) model	Preclinical digital twin-oriented simulation	Mouse model data, published pharmacokinetic parameters	Treatment optimization/Personalized therapy	PK model integration with brain network simulation explored how dopamine-targeting therapy affects M1 pathophysiology. The coupled model advances toward patient-usable digital twins for PD.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[50]	Italy	To extend and validate an agent-based digital twin model of multiple sclerosis with new treatment layers for cladribine and ocrelizumab	Computational modeling/retrospective validation	Multiple Sclerosis	Agent-based model (UISS-MS)	Prototype agent-based digital twin	Clinical trial data for cladribine and ocrelizumab (MRI and clinical outcomes)	Treatment response prediction/Personalized therapy	UISS-MS successfully replicated clinical trial outcomes for two treatments. The model was validated as a digital twin solution for predicting individual therapeutic response.	Low
[30]	Netherlands (multi-country consortium)	To establish the GEMINI consortium digital twin framework for ischemic and hemorrhagic stroke patients to improve treatment and trial design	Peer-reviewed conference proceeding/framework study	Stroke (Ischemic and Hemorrhagic)	Hybrid mechanistic model (cerebral blood flow, perfusion, thrombosis)	Framework-level digital twin	Multi-institutional patient data (12 countries, 19 organizations)	Clinical trial design/Patient-specific treatment selection	GEMINI established a multi-scale stroke digital twin spanning physiology to functional outcomes. Patient-specific digital twins will be evaluated in a prospective clinical trial.	Medium
[51]	Australia	To evaluate virtual twin-based model-informed precision dosing (MIPD-VT) for clozapine in treatment-resistant schizophrenia patients	Retrospective clinical study	Schizophrenia (treatment-resistant)	Pharmacokinetic/QSP virtual twin model (Simcyp)	Prototype virtual twin for precision dosing	Clinical PK data from hospitalized patients (N = 11)	Precision dosing/Pharmacotherapy optimization	MIPD-VT predicted steady-state clozapine concentrations with R^2 0.29–0.60. The virtual twin approach enables personalized clozapine dosing to achieve target concentrations of 350–600 ng/mL.	Low
[52]	USA	To evaluate AI-generated digital twins for enhancing clinical trial power and reducing sample sizes in ALS disease-modifying therapy trials	Retrospective clinical trial analysis	ALS (Amyotrophic Lateral Sclerosis)	Generative AI/Machine learning model	Partial digital twin for trial augmentation	Clinical trial data (10,000 + ALS participants from RCT control groups and observational studies)	Clinical trial optimization	AI-generated digital twins allowed enrollment of 18% fewer participants without compromising the study's original statistical power. The prognostic covariate-adjustment approach used in the analysis has received a qualification opinion from the EMA.	Low

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[53]	USA	To investigate how plasma NfL biomarker data can enhance digital twin-based power gains in ALS clinical trials	Retrospective cohort study	ALS (Amyotrophic Lateral Sclerosis)	Generative machine learning model	Partial digital twin for trial augmentation	Registry data (APST Research), Phase 3 clinical trial data (CEFTRIAXONE, $N = 513$), longitudinal plasma NfL measurements	Clinical trial optimization	Combining plasma NfL with digital twins increased ALSFRS-R power by up to 9% in ALS trials. Digital twins added 7% power gain even without plasma NfL measurements.	Low
[54]	USA	To develop single-cell digital twins for identifying drug targets and repurposable medicines in Alzheimer's disease	Computational modeling/in silico study	Alzheimer's Disease	Single-cell AI/machine learning model	Digital twin-adjacent molecular disease model	Single-cell transcriptomic data	Drug target identification/Drug repurposing	Single-cell digital twins characterized AD heterogeneities and identified potential FDA-approved drug candidates. The approach enables dynamic model updates from real-time feedback.	Medium
[55]	USA	To develop a digital twin approach for integrating conflicting data sources into predictive risk and resilience insights for ADRD	Computational modeling study	Alzheimer's Disease and Related Dementias (ADRD)	Bayesian computational model	Partial/digital twin-adjacent decision-support model	Literature-curated genetic, lifestyle, and environmental risk factor data	Risk prediction/Personalized prevention	Digital twins reconciled conflicting evidence on ADRD risk factors into coherent individual predictions. The framework accounts for variations in study design and baseline age.	Medium
[56]	USA	To identify causal genetic pathways of Alzheimer's disease through in silico experiments in causal AI-based digital twins	Computational modeling/in silico study	Alzheimer's Disease	Causal AI/Bayesian network model	Partial causal AI digital twin	Multiomics data (ADNI, ANMerge; ~59 K and ~35 K variables; $N = 317$ and $N = 199$)	Drug target identification/Disease mechanism understanding	41 robustly validated AD-driver genes were identified across two independent digital twins. Enriched pathways included immune response, RNA splicing, and oxidative stress.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[57]	USA	To investigate the ATN (Amyloid, Tau, Neurodegeneration) framework and its causal genetic drivers in Alzheimer's disease using digital twins	Computational modeling/in silico study	Alzheimer's Disease	Causal AI/Bayesian network model	Partial causal AI digital twin	Multiomics data (ADNI; ~59 K variables; N = 317)	Disease mechanism understanding/Precision medicine	228 ATN gene-drivers were identified; Tau abnormalities appear as early events in AD progression. Age-specific gene effects revealed younger patients have opposite directionality of Tau gene drivers.	Medium
[58]	USA	To identify causal relationships between blood metabolites, cognitive outcomes, brain imaging, and CSF biomarkers in Alzheimer's disease using digital twins	Computational modeling/in silico study	Alzheimer's Disease	Causal AI/Bayesian network model	Partial causal AI digital twin	Multiomics data (ADNI; ~59 K clinical and molecular variables)	Biomarker discovery/Disease mechanism understanding	Causal metabolites actively driving AD were distinguished from that merely reflecting disease progression. Findings provide new insights into AD pathogenesis for therapeutic targeting.	Medium
[59]	Switzerland	To develop a wearable sensor-driven musculoskeletal digital twin for quantitative gait biomechanics assessment in post-stroke individuals	Proof-of-concept/validation study	Stroke (post-stroke rehabilitation)	EMG-driven musculoskeletal simulation model	Prototype sensor-driven digital twin	Wearable sensor data (EMG, IMU) from post-stroke individuals	Rehabilitation monitoring/Motor impairment assessment	The wearable digital twin estimated ankle torques with R^2=0.65 ± 0.21. EMG signals compensated for IMU inaccuracies, enabling out-of-lab biomechanical assessment.	Low
[60]	USA	To introduce and validate a digital twin brain framework linking individual connectomes to multitask neurobehavioral predictions for personalized therapy	Prospective computational modeling study	Psychiatric/neurological disorders (general brain function)	Hypernetwork-based recurrent neural network	Partial personalized in silico intervention model	Resting-state connectome and task-based fMRI data (N = 228 individuals)	Personalized therapy/In silico intervention	The model predicted behavioral choices with > 90% accuracy and BOLD patterns (r = 0.84). In silico interventions successfully modulated targeted cognitive and affective functions.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[61]	UK	To develop a blockchain-enabled, ML-powered digital twin application for brain stroke prediction with enhanced security and scalability	Proof-of-concept/technical study	Stroke	Machine learning model with consortium blockchain integration (Logistic Regression + Ganache)	Digital twin-adjacent secure prediction framework	Public Kaggle brain stroke dataset (N = 4,981); synthetic patient EHR, supplementary, and transactional data files	Stroke prediction/ Secure health data management	The blockchain-enabled digital twin achieved 98.28% stroke prediction accuracy, exceeding prior literature (84–92%). Consortium blockchain with smart contracts successfully detected and reverted tampered patient data in real time.	Medium
[62]	USA	To evaluate digital twins for Alzheimer's disease using data from a completed Phase 2 clinical trial	Retrospective clinical trial analysis	Alzheimer's Disease	Deep learning model	Partial digital twin for clinical trial optimization	Phase 2 clinical trial data (crenezumab ABBY study)	Clinical trial optimization	Digital twins reduced required trial subjects while meeting regulatory requirements. Historical control data was leveraged to accelerate RCT design without sacrificing reliability.	Medium

Table 2 (continued)

References	Country	Aim	Study design	Neurological condition	Digital twin approach	Operational digital twin classification	Data sources	Application domain	Key findings	Methodological appraisal rating
[63]	Australia	To develop and test a digital twin discovery algorithm for matching phases of cognitive decline in dementia patients	Proof-of-concept study (synthetic data)	Dementia	Data-matching algorithm (MAPE-based twin discovery)	Digital twin-adjacent twin-discovery model	Synthetic MMSE score data (30 past + 5 present subjects)	Disease progression modeling/ Clinical decision support	Digital twin segments with MAPE < 2% were identifiable for all 5 present subjects. Results support generalizability to multi-dimensional clinical data.	Medium
[64]	China	To construct digital twin brain models for prognosis prediction in patients with disorders of consciousness	Retrospective computational study	Disorders of Consciousness (DOC)	Mathematical brain model + machine learning (SVM classifier)	Partial/digital twin-adjacent prognostic model	EEG data from DOC patients with different outcomes	Prognosis prediction	Digital twin brain model parameters with SVM classifier predicted DOC prognosis outcomes. The approach provided insights into recovery mechanisms beyond traditional EEG features.	Medium

ALS, amyotrophic lateral sclerosis; AD, Alzheimer's disease; ADRD, Alzheimer's disease and related dementias; AF, atrial fibrillation; AI, artificial intelligence; AE, autoencoder; ANN, artificial neural network; BOLD, blood oxygen level–dependent; CFD, computational fluid dynamics; CNN, convolutional neural network; CT, computed tomography; DOC, disorders of consciousness; DT, digital twin; EEG, electroencephalography; EHR, electronic health record; EMG, electromyography; FDA, Food and Drug Administration; fMRI, functional magnetic resonance imaging; HDT, human digital twin; ICD, International Classification of Diseases; IoT, Internet of Things; IMU, inertial measurement unit; KNN, k-nearest neighbors; LA, left atrium; MAS, Modified Ashworth Scale; MEG, magnetoencephalography; ML, machine learning; MMSE, Mini-Mental State Examination; MRI, magnetic resonance imaging; MS, multiple sclerosis; PD, Parkinson's disease; PDE, partial differential equation; PK, pharmacokinetics; QSP, quantitative systems pharmacology; RCT, randomized controlled trial; RF, radio frequency; RNN, recurrent neural network; SAE, stacked autoencoder; SARA, Scale for the Assessment and Rating of Ataxia; SCD, subjective cognitive decline; SVM, support vector machine; UWB, ultra-wideband. Methodological appraisal rating: Low = clearly specified digital twin or virtual modeling architecture, explicit data sources and input variables, quantitative validation or structured clinical/implementation evaluation, and sufficient methodological detail for interpretation. Medium = relevant digital twin or digital twin–adjacent model with adequate description but evidence primarily based on retrospective, simulation-based, proof-of-concept, conceptual, or internally validated work without external validation, prospective validation, or clinical implementation. Ratings were used for descriptive synthesis only and did not influence study inclusion

(22.2%) [33, 36, 41, 44, 56–58, 63] and 6 studies (16.7%) [34, 38, 40, 44, 55, 61], respectively. Treatment optimization or personalized therapy featured in seven studies (19.4%) [39, 46, 49–51, 54, 60]. Rehabilitation monitoring and motor function assessment were the focus of four studies (11.1%) [42, 48, 59, 60], while early diagnosis and biomarker discovery were each addressed in three studies (8.3%) [17, 54, 56] and [17, 41, 58].

Based on the qualitative methodological appraisal, 27 studies (75.0%) received a medium rating, and 9 studies (25.0%) received a low rating [17, 37, 38, 43, 50–53, 59]. The ratings indicate that most studies provided adequate model descriptions but relied on retrospective, simulation-based, proof-of-concept, conceptual, or internally validated evidence rather than on prospective or external validation. Taken together, these findings suggest that although the literature is growing rapidly, most studies remain in intermediate stages of methodological maturity.

3.3 Classification of digital twin approaches

The digital twin approaches across the 36 included studies were classified into eight categories based on the primary modeling paradigm: (i) data-driven and machine learning models; (ii) mechanistic and physics-based models; (iii) hybrid mechanistic–data-driven models; (iv) pharmacological and quantitative systems pharmacology (QSP) models; (v) neurophysiological and brain simulation models; (vi) wearable and sensor-driven models; (vii) human digital twin and rehabilitation frameworks; and (viii) expert and knowledge-based models (Table 3). In addition to the modeling paradigm, the included studies were interpreted in terms of digital twin maturity. Most studies presented prototypes or partial digital twins rather than fully dynamic ones. Several studies proposed conceptual frameworks or digital twin–adjacent enabling models, including sensing, secure data management, or static prediction approaches relevant to future digital twin development. Therefore, the term “digital twin” in this review should be interpreted as a maturity spectrum rather than as a uniform indication of fully dynamic, continuously updated, bidirectional digital twin systems.

Data-driven and machine learning models constituted the largest category (n = 16 studies). Subtypes included generative AI models (variational autoencoders, deep generative models) applied to ALS, Parkinson’s disease, Alzheimer’s disease, and stroke [31–33, 35, 52–55, 62]; causal AI and Bayesian network models applied to Alzheimer’s disease and related dementias [34, 55–58]; deep learning architectures (autoencoders, RNNs, CNNs, hypernetworks) applied to stroke and psychiatric brain function [47, 60–62]; single-cell transcriptomic digital twins for Alzheimer’s disease [54]; and statistical or interpretable machine learning models for Alzheimer’s disease and multiple sclerosis [32, 36, 45].

Comparative synthesis across modeling paradigms highlighted several architecture-level trade-offs. Data-driven and machine learning models showed high scalability and flexibility for large clinical, imaging, and trial datasets, but validation was mostly retrospective, and mechanistic interpretability was often limited. Mechanistic and physics-based models offered stronger biological plausibility and clearer explanatory structure, particularly for brain network dynamics, tumor growth, immune simulation, and hemodynamic modeling, but required intensive parameterization and showed limited scalability across heterogeneous patient populations. Hybrid mechanistic–data-driven models provided a potential balance between predictive performance and physiological interpretability, although evidence remained limited to proof-of-concept or framework-level studies. QSP and pharmacokinetic virtual twin models demonstrated relatively high translational relevance for treatment optimization and in silico trial augmentation, particularly for rare or treatment-resistant conditions. Wearable, sensor-driven, and rehabilitation-oriented human digital twin models appeared closer to clinical implementation because of continuous monitoring and adaptive feedback potential; however, external validation, workflow integration, and long-term outcome evidence remained limited. Overall, no single architecture demonstrated superiority across validation, interpretability, scalability, and clinical readiness, suggesting that model selection should depend on clinical purpose, data availability, and implementation context.

Mechanistic and physics-based models (n = 5) comprised ODE/PDE brain network and tumor growth models for glioblastoma, Parkinson’s disease, and epilepsy [39, 43, 49]; an agent-based immune simulation for multiple sclerosis [50]; and computational fluid dynamics models of atrial fibrillation-related stroke risk [38, 40]. Hybrid mechanistic–data-driven models (n = 2) combined physics-informed simulation with machine learning for stroke, including the multinational GEMINI consortium framework [30, 44].

QSP models applied PBPK and Simcyp-based virtual twin platforms to treatment-resistant schizophrenia [51] and Pompe disease [46]. Neurophysiological brain simulation models integrated EEG/MEG data with computational network architectures for disorders of consciousness, epilepsy, and preclinical Alzheimer’s disease [17, 39, 64]. Wearable and sensor-driven models utilized EMG, RF/UWB, and IoT data for post-stroke rehabilitation, stroke detection, and secure health monitoring [47, 59, 61]. Human digital twin and rehabilitation frameworks supported adaptive therapy delivery in upper-limb stroke rehabilitation and general

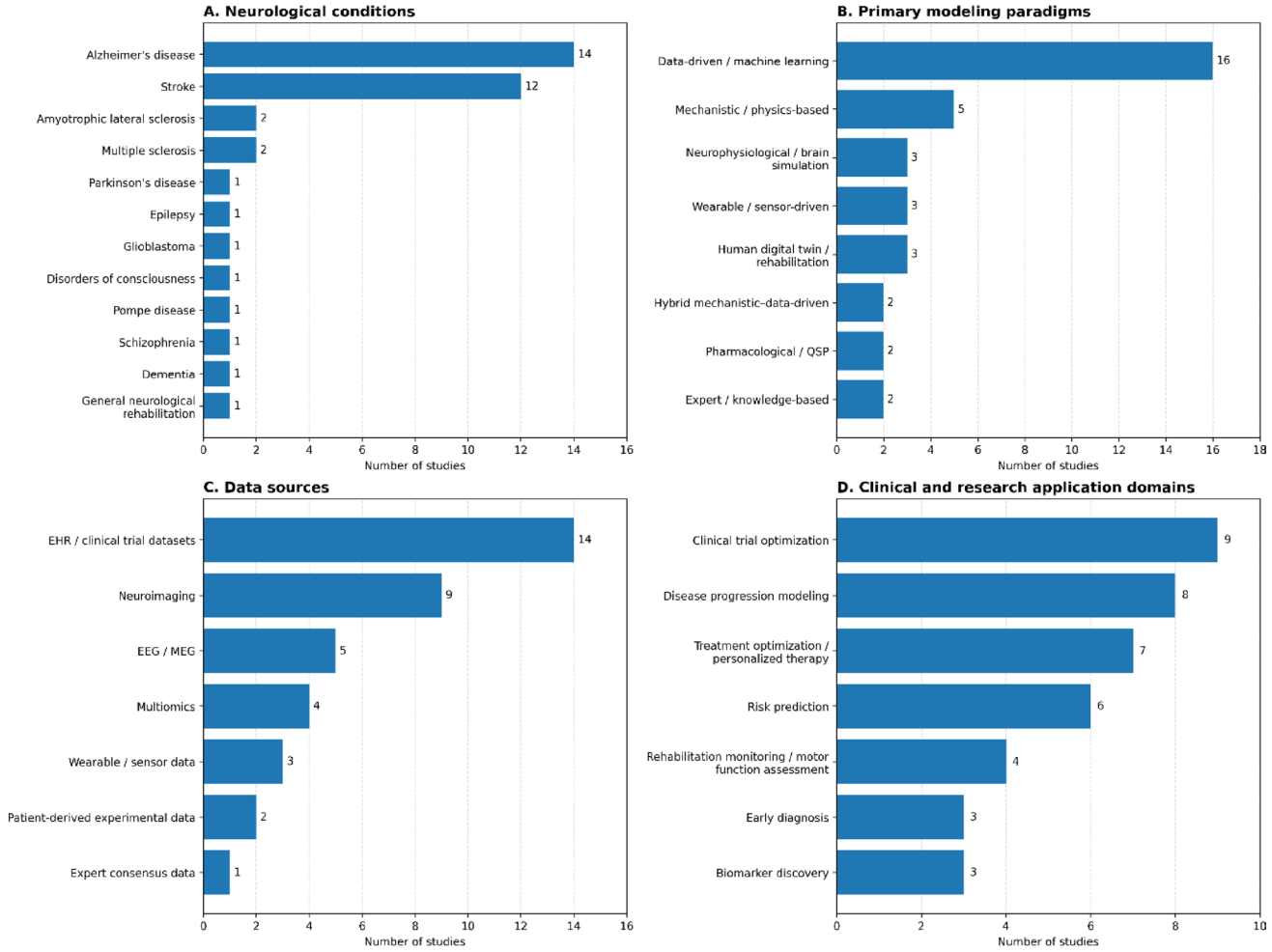


Fig. 2 Distributional patterns of the included studies according to **A** neurological condition, **B** primary modeling paradigm, **C** data source, and **D** clinical or research application domain. Counts are not mutually exclusive because some studies addressed multiple neurological conditions, data modalities, or application domains

neurological rehabilitation [35, 42, 48]. Expert and knowledge-based models, including a Delphi consensus rule-based digital twin for acute stroke [37] and a MAPE-based data-matching algorithm for dementia [63], completed the classification.

Across all categories, Alzheimer's disease and stroke collectively accounted for 22 of 36 studies (61.1%) of digital twin applications. Clinical trial optimization predominated in Alzheimer's disease and ALS studies, whereas rehabilitation monitoring and disease progression modeling were the primary application domains for stroke and multiple sclerosis.

4 Discussion

4.1 Principal findings

This scoping review systematically mapped 36 studies examining digital twin applications across neurological conditions, demonstrating a rapid increase in publications since 2023 and a predominance of research originating from predominantly from North America and Europe, with the United States contributing the largest share. Alzheimer's disease and stroke accounted for the majority of applications, collectively representing approximately two-thirds (61.1%) of included studies. Most studies employed data-driven or machine learning–based approaches, followed by mechanistic and hybrid models, reflecting methodological diversity in this emerging field. Clinical trial optimization, disease progression modeling, and risk prediction were the most common application domains, whereas real-world clinical implementation and randomized trials were notably

Table 3 Classification of digital twin approaches in neurological conditions

Category	Approach/model type	Neurological conditions	References
Data-driven/machine learning	Generative AI (Variational/Deep generative models)	ALS; Parkinson's disease; Alzheimer's disease; Stroke	[31– [33, 52–53, 62]
	Causal AI/Bayesian Network (Probabilistic graphical models)	Alzheimer's disease; ADRD	[34, 55–58]
	Deep Learning/Neural Networks (AE, RNN, CNN, Hypernetwork)	Stroke; Alzheimer's disease; Psychiatric/Brain function	[47, 60–62]
	Single-Cell AI (Transcriptomic digital twins)	Alzheimer's disease	[54]
	Statistical/Interpretable ML (Mixed models, spline, prediction-powered inference)	Alzheimer's disease; Multiple Sclerosis	[32, 36, 45]
Mechanistic/physics-based	Mathematical/ODE/PDE Models (Tumor growth, brain network dynamics)	Glioblastoma; Parkinson's disease; Epilepsy	[39, 43, 49]
	Agent-Based Model (Immune system simulation)	Multiple Sclerosis	[50]
	Computational Fluid Dynamics (CFD) (Hemodynamic/cardiac simulation)	Stroke (AF-related) (Thrombosis/stroke risk)	[38, 40]
	Systems Biology/Homeostatic Model (Multi-process mechanistic simulation)	Alzheimer's disease	[41]
Hybrid (mechanistic + data-driven)	Hybrid Mechanistic + ML (Physics-informed learning)	Stroke; Stroke (risk)	[38, 44]
	Multi-Scale Consortium Framework (Physiology → functional outcome)	Stroke (ischemic + hemorrhagic)	[30]
Pharmacological/quantitative systems pharmacology (QSP)	"QSP/Pharmacokinetic Virtual Twin (PBPK/Simcyp)"	Schizophrenia (treatment-resistant); Pompe disease (neuromuscular)	[46, 51]
Neurophysiological/brain simulation	EEG-Based Brain Network Model (Mathematical + SVM classifier)	Disorders of consciousness; Epilepsy	[39, 64]
	Neurophysiological + Computational Brain (EEG/MEG-driven twin)	Alzheimer's disease (preclinical/SCD)	[17]
Wearable/sensor-driven	EMG-Driven Musculoskeletal Model (Biomechanical simulation)	Stroke (post-stroke rehabilitation)	[59]
	RF/UWB Sensing + ML (Backscattering-based detection)	Stroke	[47]
	IoT/Blockchain-Enabled DT (Secure health monitoring + ML prediction)	Stroke	[61]
Human digital twin (HDT)/rehabilitation	HDT with Exoskeleton + Serious Games (Adaptive therapy platform)	Stroke (upper-limb rehabilitation)	[48]
	ICF-Based Integrated DT (Therapy device data + outcome measures)	Neurological disorders (general rehabilitation)	[42]
	Patient-Specific Updating Model (Continually updating predictive twin)	Stroke (post-stroke aphasia)	[35]
Expert/knowledge-based	Delphi Consensus Rule-Based DT (Expert-derived clinical rules)	Stroke (acute)	[37]
	Data-Matching Twin Discovery (MAPE-based historical patient matching)	Dementia	[63]

ALS, amyotrophic lateral sclerosis; AD, Alzheimer's disease; ADRD, Alzheimer's disease and related dementias; AF, atrial fibrillation; CFD, computational fluid dynamics; DOC, disorders of consciousness; DT, digital twin; EEG, electroencephalography; EHR, electronic health record; EMG, electromyography; HDT, human digital twin; ICF, International Classification of Functioning; IMU, inertial measurement unit; IoT, Internet of Things; ML, machine learning; MS, multiple sclerosis; PD, Parkinson's disease; PK, pharmacokinetic; QSP, quantitative systems pharmacology; RCT, randomized controlled trial; RF, radio frequency; SCD, subjective cognitive decline; UWB, ultra-wideband

absent. Overall, the evidence base remains at an early stage of translational maturity, with 27 studies (75.0%) receiving a medium methodological appraisal rating and 9 studies (25.0%) receiving a low rating, and with limited prospective validation or clinical integration.

4.2 Comparison with literature

The present findings are highly consistent with the broader digital twin and precision medicine literature, which characterizes the field as rapidly expanding yet still predominantly preclinical or translational in nature [31, 56, 66]. Prior reviews have emphasized that digital twin applications are disproportionately concentrated in diseases with high societal burden and rich multimodal datasets, such as Alzheimer's disease and stroke, due to the availability of neuroimaging, biomarker, and longitudinal cohort data [67–69]. The current review confirms this trend, with these two conditions accounting for the majority of identified studies.

The predominance of machine learning–based approaches aligns with recent developments in biomedical AI, where deep learning, generative models, and causal inference frameworks have become central tools for modeling complex disease systems [41, 62, 70]. In particular, generative AI–based digital twins have been increasingly applied to simulate disease trajectories and augment clinical trial data, reflecting a shift toward data-centric modeling paradigms [62, 66, 71]. However, this trend also highlights a divergence from earlier conceptualizations of digital twins, which emphasized mechanistic fidelity and physiological realism as defining characteristics [7, 9]. The architecture-level synthesis also indicates that digital twin development in neurology involves a trade-off between predictive scalability, mechanistic interpretability, and clinical readiness. Data-driven approaches can support large-scale prediction, trial augmentation, and disease trajectory modeling, but limited transparency may restrict clinical trust and regulatory acceptance. Mechanistic approaches provide stronger explanatory value and biological plausibility, but broad implementation requires detailed calibration, specialized data, and disease-specific assumptions. Hybrid models may offer a promising direction for precision neurology by combining predictive learning with physiological constraints; however, current evidence remains insufficient for routine clinical deployment. Sensor-driven and rehabilitation-oriented human digital twins appear relatively closer to clinical translation because continuous monitoring and adaptive feedback align with real-world care processes, but prospective validation and workflow integration remain underdeveloped. Therefore, future digital twin research should avoid treating architectures as interchangeable and should instead align model design with intended use, data infrastructure, interpretability needs, and implementation context.

Compared with cardiology and oncology, where digital twin models have progressed toward clinical prototyping and early implementation, neurological applications appear relatively less mature [7, 20]. This discrepancy may be attributed to the brain's intrinsic complexity, challenges in obtaining high-resolution longitudinal data, and the limited availability of validated biomarkers for many neurological disorders [30, 72, 73]. The maturity classification in this review supports this interpretation by showing that neurological digital twin research remains concentrated in prototype, partial, conceptual, and digital twin–adjacent models rather than in clinically implemented systems. The lack of randomized controlled trials and prospective validation is also consistent with prior reports identifying regulatory, ethical, and methodological barriers as key obstacles to clinical translation [28, 69].

4.3 Mechanisms

First, the observed patterns in digital twin applications illustrate several underlying mechanisms grounded in computational neuroscience, systems biology, and AI-driven modeling. Neurological diseases exhibit multiscale complexity, with interactions across genetic, molecular, cellular, and network levels. This complexity necessitates modeling approaches capable of integrating heterogeneous data types, including neuroimaging, electrophysiological signals, omics data, and clinical variables. Machine learning models, particularly deep learning architectures, are well-suited to capturing high-dimensional, nonlinear relationships, which explains widespread adoption in the included studies.

Second, digital twins enable in silico experimentation by simulating individualized disease trajectories and treatment responses. This capability is particularly valuable in neurology, where disease progression is often slow, heterogeneous, and difficult to study using traditional clinical trial designs. Virtual clinical trials and synthetic control arms generated from digital twins can enhance statistical power, reduce sample sizes, and accelerate therapeutic development [46, 62, 71]. These mechanisms are especially relevant in rare neurological diseases and conditions such as ALS, where patient recruitment is challenging.

Third, mechanistic and hybrid models provide critical advantages by incorporating biological plausibility and causal reasoning. Mechanistic models, including differential equation–based and agent-based frameworks, can simulate underlying disease processes such as neuronal network dysfunction, protein aggregation, and neuroinflammation [39, 43, 50]. Hybrid models that combine mechanistic simulations with machine learning offer a promising approach to balance predictive performance with interpretability, enabling more clinically actionable insights [30, 44]. However, the relatively limited number of such models identified in this review suggests ongoing technical and methodological challenges in integrating these approaches at scale.

Temporal modeling is another important mechanism for neurological digital twins because disease progression, treatment response, and functional recovery often unfold over extended periods. Longitudinal data integration can support iterative refinement of individualized simulations and may improve the prediction of changing disease states. In this context, temporal modeling should be considered a core design requirement for future neurological digital twins, particularly for neurodegenerative disease, epilepsy monitoring, stroke recovery, and rehabilitation applications.

Finally, data infrastructure and interoperability are central to enabling digital twin development. The integration of diverse data sources, including electronic health records, wearable devices, and imaging systems, requires

standardized data pipelines and robust computational frameworks [27, 72]. The absence of such infrastructure in many healthcare settings may partially explain the limited number of clinically implemented digital twin models observed in this review.

4.4 Clinical implications

The findings of this review underscore the transformative potential of digital twin technologies in precision neurology. One of the most immediate applications lies in clinical trial optimization, where digital twins can function as synthetic control arms or augment existing trial data. This approach has been shown to reduce required sample sizes, improve statistical efficiency, and address ethical concerns related to placebo use, particularly in progressive neurological diseases [62, 71]. Such innovations may significantly accelerate drug development pipelines and reduce associated costs.

In clinical practice, digital twins offer the potential to enhance early diagnosis and risk stratification. By integrating multimodal data, including neuroimaging, biomarkers, and digital phenotyping, digital twins can identify individuals at high risk of disease progression before the onset of overt clinical symptoms [11, 17]. This capability is particularly relevant for neurodegenerative diseases such as Alzheimer's disease, where early intervention may have the greatest impact on disease trajectory.

Digital twins also have important implications for personalized treatment and decision support. By simulating individual responses to different therapeutic interventions, digital twins can help clinicians select optimal treatment strategies tailored to patient-specific characteristics [35, 51]. In addition, rehabilitation-focused digital twins incorporating wearable sensors and human digital twin frameworks can enable continuous monitoring and adaptive therapy, improving functional outcomes in conditions such as stroke [48, 59].

Despite these promising applications, several barriers must be addressed before widespread clinical adoption can be achieved. These include the need for rigorous external validation, standardized evaluation metrics, and regulatory approval pathways [29, 69, 73]. Ethical considerations, including data privacy, algorithmic bias, and accountability, are particularly critical in neurological applications, where clinical decisions may have profound implications for patient autonomy and quality of life [28, 65]. Furthermore, integration into clinical workflows and clinician acceptance remain key challenges that must be addressed through user-centered design and interdisciplinary collaboration.

4.5 Strengths and limitations

This review provides a comprehensive synthesis of digital twin applications across neurological conditions, capturing methodological diversity and emerging trends in a rapidly evolving field. The use of a systematic search strategy across multiple databases and adherence to PRISMA-ScR guidelines enhances the rigor and transparency of the findings. Additionally, classifying digital twin approaches into distinct categories provides a valuable framework for understanding current research directions and identifying gaps.

However, several limitations should be considered. First, the restriction to English-language publications may have introduced selection bias and limited the inclusion of relevant studies from non-English-speaking regions. Second, as a scoping review, the objective was to map the evidence rather than to assess effectiveness; therefore, no quantitative synthesis or meta-analysis was performed. Third, substantial heterogeneity in study designs, data sources, digital twin architectures, and outcome measures limits comparability across studies and precludes definitive conclusions about clinical effectiveness. A related limitation concerns conceptual heterogeneity and translational maturity. Although operational criteria were used to distinguish digital twins from broader computational modeling approaches, the included literature spanned prototype, partial, conceptual, and digital twin–adjacent models. As a result, this review can map technology development and methodological trends but cannot establish clinical effectiveness or implementation readiness. Fourth, publication bias may be present, as studies with positive findings are more likely to be published. Finally, the rapidly evolving nature of digital twin technologies means that recent advances may not have been fully captured within the search timeframe, highlighting the need for continuous updates to the evidence base.

4.6 Future research

Future research should prioritize prospective validation, external evaluation, and clinical implementation studies of digital twin–based systems. Randomized controlled trials and pragmatic implementation studies are essential to establish clinical effectiveness, safety, and cost-effectiveness [29, 71]. In particular, hybrid modeling approaches that integrate mechanistic and data-driven methods should be further developed to enhance both predictive accuracy and interpretability [44]. Standardization efforts are also needed, including clear definitions of digital twins, standardized reporting guidelines, and consensus on evaluation metrics. Interoperable data infrastructures capable of integrating electronic health records, imaging data, and wearable sensor

data will be critical to supporting dynamic and clinically useful digital twin systems [27, 72].

Future research should also address ethical and regulatory challenges, including data privacy, algorithmic bias, and accountability frameworks. Collaboration between clinicians, engineers, data scientists, and policymakers will be essential to navigate these complex issues and facilitate safe clinical implementation [28].

Finally, expanding research to underrepresented neurological conditions and diverse populations will be important for ensuring equitable access to digital twin technologies. Longitudinal cohort studies with diverse datasets will improve generalizability and support the development of robust, inclusive digital twin models.

5 Conclusion

This scoping review demonstrates that digital twin applications in neurological conditions are rapidly increasing but remain largely at an early, preclinical stage. Current research is dominated by data-driven and machine learning–based approaches, with key applications in clinical trial optimization, disease progression modeling, risk prediction, and personalized treatment. These findings highlight the growing role of digital twins as a potential enabler of precision neurology. However, current evidence remains limited by sparse prospective validation, limited external evaluation, and the absence of randomized controlled trials. Future research should focus on standardized frameworks, multimodal real-world data integration, clinical validation, and implementation pathways. Addressing regulatory and ethical challenges will also be essential. With continued advancement, digital twin technologies have strong potential to improve clinical decision-making and transform the management of neurological diseases.

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Declarations

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